

New Swift AI Software Can Help Transform Address Structuring

Features

Open source and free of charge

Trained on global Swift payment dataset

Offline operation; no external API sensitive data sharing

Customisable for specific geographies and postal address needs

Inferences as an option

Benefits

Universal applicability

Adaptable to your needs without licensing costs

Guaranteed privacy and compliance, trained on synthetic data

Evaluated by 3rd party for trust and credibility

Supports compliance

Transparency and auditability with supporting diagnostic data.

Address structuring requirements

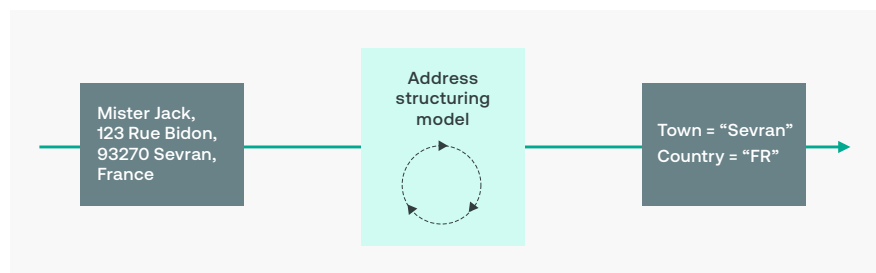
Starting November 2026, the Swift network will no longer accept unstructured postal addresses in CBPR+ ISO 20022 messages.

Swift customers and participants in market infrastructures that align to the HVPS+ will need to complete a mandatory transition from free-text address fields to the structured ISO 20022 format. Only fully structured or hybrid addresses will be allowed, with mandatory presence for Town and Country fields, or risk having failed messages.

Structured addresses are vital for financial institutions as they ensure higher data integrity, streamline payment processing, and strengthen adherence to regulatory requirements.

How Swift can support you

Swift's Natural Language Processing based AI solution enables the automatic extraction of mandated structured address elements - specifically Town and Country - from legacy unstructured data that can still be found in corporate and client databases, or in payments files received through corporate channels. It provides high-confidence predictions for the countries and towns found in the free format content, including confidence scores and multiple resolution suggestions for manual or automated review. The solution leverages network-wide information spanning more than 200 countries and territories, ensuring comprehensive geographic coverage and has been successfully tested with pilot customers.



Open source

To help the community meet ISO 20022 postal address migration deadlines, Swift is - for the first time - delivering an open-source AI solution, made available free of charge for our Swift community. This reflects our commitment to collaboration, innovation, and the long-term success of the ecosystem. The model is accessible to the Swift

community (e.g. Swift users, service bureaus, vendors and Swift partners) for everyone to use and build on. Integration is flexible - it can be integrated into internal systems, standalone tools, or used as a batch-processing engine to pre-process and clean unstructured address fields in legacy message formats.

How does it work?

Step 1: Extraction of Town and Country entities from address strings

The input to the model consists of tokenised components of the address strings typically contained in debtor and creditor fields, potentially enriched with linguistic and contextual features.

To support accurate extraction, the model leverages external data sources such as global address datasets, postal code registries, and open-source geolocation databases.

For financial institutions operating across multiple jurisdictions with unique requirements, the address structuring model can be adjusted by incorporating proprietary or additional registries and repositories tailored to specific regions.

Privacy and compliance

The model was trained on synthetic data generated from templates that reflect the diversity of address structures found in real Swift transactions (including edge cases and exceptions) in more than 200 countries and jurisdictions.

This means it contains no personal data and there is no risk of sensitive data leakage. The software is also self-contained. The customer downloads it, and it runs fully offline and isolated within their systems; there is no need to submit API calls containing sensitive data to external parties.

For more information, contact your Swift account manager or email us at swift.address structuring@swift.com

Disclaimer

This software is provided as is, without any promises or guarantees. That means that although we have done our best to make it accurate and reliable, we don't guarantee that it will always work well, meet your needs, or be free from problems.

By using it, you accept that Swift isn't responsible for anything that goes wrong – including bugs, or any other issues. Except for security issues, which Swift will address if reported.

Swift is a member-owned cooperative, providing secure financial messaging services to more than 11,000 organisations, across the financial ecosystem, in almost every country in the world. For nearly five decades we have delivered certainty, continuity and excellence by constantly evolving in an ever-changing landscape. In today's fast moving, increasingly connected and challenging world, this approach has never been more relevant.

swift.com

Step 2: Prediction of Town and Country

The output is limited to Town and Country predictions. Confidence scores help customers assess the accuracy and reliability of Town and Country extraction.

Based on your risk appetite, the model can be further tuned to switch on its inference parameters to allow further enrichment based on calculated confidence criteria.

“As we progress with ISO 20022 migration, our next major challenge is the transition to structured address requirements. By November 2026, free-text address fields must be replaced with structured or hybrid ISO format. At Société Générale, we are fully committed to meeting these deadlines. However, because party addresses are captured at initiation, often outside our control, unstructured data can still lead to manual intervention, delayed payments, and higher compliance risk. Swift's address structuring capabilities are interesting to ensure mandatory components such as Town and Country are included, upfront or on an ongoing basis, thereby reducing errors and improving straight-through processing.”

Laurence DORE, ISO 20022 Program Direction, Société Générale
